Classical Systems Biology

MCSB Bootcamp — Mathematical and Computational Track

Jun Allard

Early synthetic biology

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LETTERS

A bottom-up approach to gene regulation

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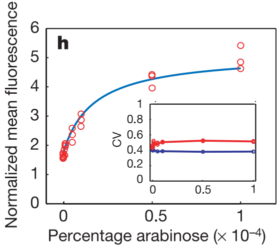
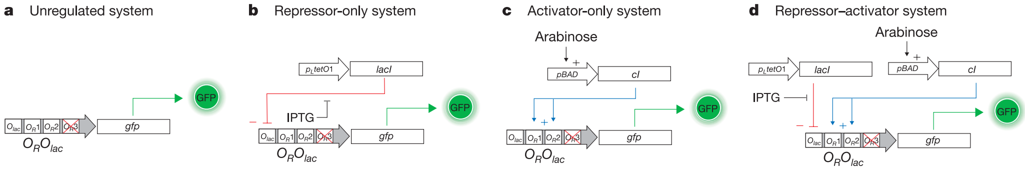
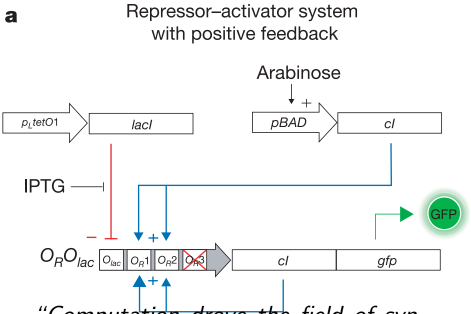




What happens when you add positive feedback?



$\rightarrow$?

"Computation drove the field of synthetic biology in its very early days."
Collins, 2020

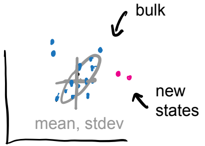
"We didn't have those data sets. We didn't have a wet lab. We didn't have the capability. We had to rely on mathematical modeling."
paraphrasing Collins, 2015

Complex systems and “systems thinking”

The goals, challenges, systems, tools and approaches of today

Data and analytics

Keystone **goal**: discover the hidden — new states, cell types, disease classes, ...



Keystone **challenge**: biological data is not just big, it is **high-dimensional**.

- 30000 genes, 1000 cells
- 10^4 microbial species, 20 patients

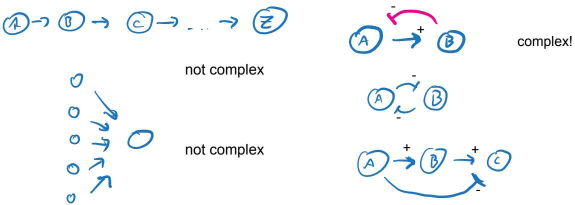
Keystone **tool**: clustering, regression, dimension-reduction ... coding.

Models and mechanism

Keystone **goal**: **prediction** — "what would happen if..."

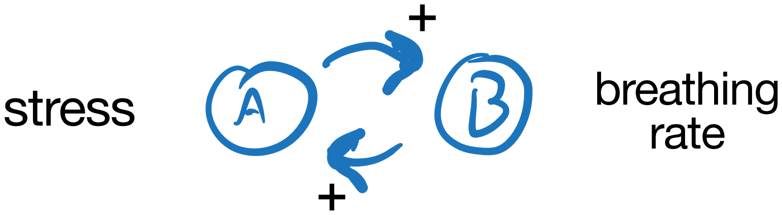
if A is increased, then...

Keystone **challenge**: biological systems are **complex**, not just big.



Keystone **tool**: dynamical systems, control theory, (the differential equation,) ...

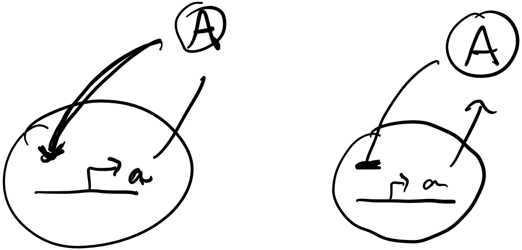
Most things in biology have feedback

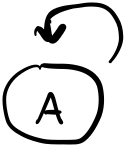


Complex biological systems and "systems thinking"

→ Activation

⊣ Repression





Positive feedback

With positive feedback, the curve showing steady-state protein as a function of activator TF should move up.

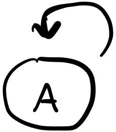
Does the curve move up by:

- shifting the EC50 to the left?
- moving up at high-C?
- moving up at low-C?
- a combination of all three?

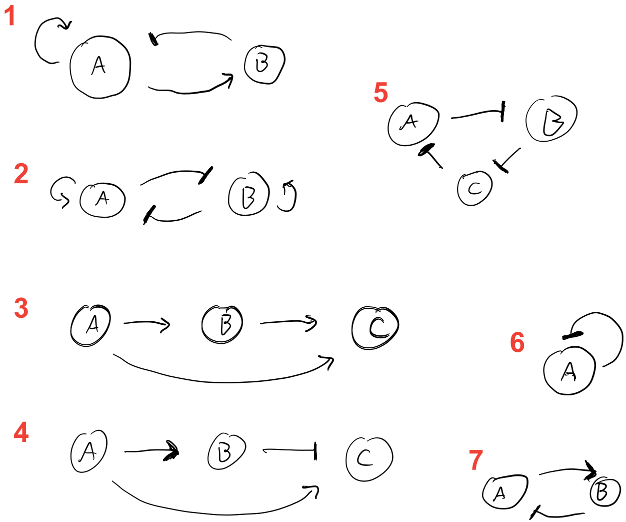
Seven circuits


Activation


Repression

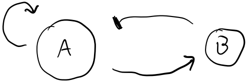


Positive feedback



In-class activity: matching

1



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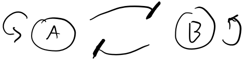

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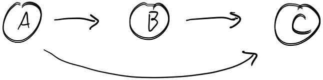

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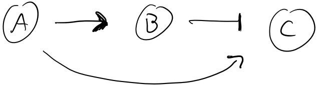

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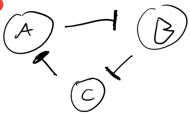

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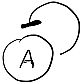

7

Match each circuit to its name.

- Direct negative feedback
- Indirect negative feedback
- Double-negative feedback
- Fast positive and slow negative feedback
- Incoherent feedforward
- Coherent feedforward
- The Repressilator